

- 2 (a) A body travelling at a constant speed in a circular path experience centripetal acceleration. Using Newton's laws of motion explain why there is acceleration although the speed is constant.

The velocity of the body changes along a circular path. By N1L this require an external resultant force to act on the body [B1]. Since the centripetal acceleration is pointing to the centre of circle and perpendicular to the instantaneous velocity, by N2L, it has no component along the path, hence speed is constant.[B1].....[2]

- (b) A car of mass 1500 kg travels in a horizontal circular path of radius 50.0 m on a banked road with speed of  $15.0 \text{ m s}^{-1}$  without any frictional force acting on the tyres along the slope.

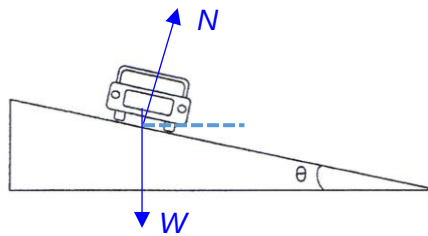


Fig. 2.1

- (i) Calculate the angle  $\theta$  at which the road is banked.

$$N \cos \theta = mg$$

$$N \sin \theta = \frac{mv^2}{r} \quad [\text{M1 both eqn}]$$

$$\tan \theta = \frac{v^2}{rg} = \frac{15^2}{50.0 \times 9.81} \quad [\text{M1}]$$

$$\theta = 24.6^\circ \quad [\text{A1}]$$

$$\theta = \dots\dots\dots^\circ \quad [3]$$

- (ii) Explain how friction force enables the car to travel in the same horizontal circular path at a lower speed.

The horizontal component of static friction acts away from the direction of centripetal force [B1], resulting in a smaller magnitude of centripetal force [B1]. A smaller centripetal force permits the car to move on the banked surface in uniform circular motion with a slower speed. ....[2]

[Total: 7]

- 3 (a) (i) The kinetic theory for an ideal gas of volume  $V$  at pressure  $p$  leads to the equation